UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

GCE Ordinary Level

MARK SCHEME for the October/November 2011 question paper for the guidance of teachers

5070 CHEMISTRY

5070/42 Paper 4 (Alternative to Practical), maximum raw mark 60

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1 **(a)** B (1)

(b) pipette (1) [2]

2 **(a)** **(i)** insoluble (in water) not slightly soluble (1)

(ii) catalyst or speeds up reaction or dehydrating agent (1)

(iii) $CH_2 = CH_2$ (1)

(b) **(i)** yellow, brown or orange (1)

(ii) colourless or decolourised (1)

(iii) addition or saturation (1)
not bromination

(c) **(i)** butene ignore 1 or 2 in name (1) $CH_3CH_2CH=CH_2$ etc (1) [8]

3 **(a)** sea water is higher as it contains impurities, or dissolved salts (which increases the boiling point) (1)

(b) NaC*l* (1)

(c) **(i)** flask containing water (by label or observation) together with heat either shown as a bunsen or heating block or arrows labelled heat (1)

condenser showing outer and inner tube, sloping in correct direction, water labelled entering and leaving at correct points, (1)
(no vertical condensers)

receiver and all the apparatus connected correctly (1)
No blockages
Ignore use of thermometer and fractionating tube

(ii) desalination or reversed osmosis (1)

(d) filtration or sedimentation or centrifugation (1)

(e) chlorine (1) bleaches or decolourises litmus (1) [9]

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4 **(a)** syringe (1)

(b) lime water turns milky (1)

(c) **(i)** 0.005 (1)

(ii) 0.01 (1)

(iii) 0.01 x 84 = 0.84 g (1) [5]

(5) (d) (1) [1]

(6) (d) (1) [1]

(7) (c) (1) [1]

(8) (b) (1) [1]

(9) (c) (1) [1]

10 **(a)** yellow to blue (1)

(b)

26.1	28.6	37.1	1 mark for each correct row or column (3)
0.0	3.4	11.7	
26.1	25.2	25.4	

Mean titre = 25.3 (1) cm^3

(c) 0.00506 / 0.0051 (1)

(d) 0.00506 / 0.0051 (1)

(e) 0.0506 / 0.051 (1)

(f) **(i)** 0.0506 / 0.051 (1)

(ii) 2.02(4) / 2.04 (1) [10]

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11 **(a)** colourless solution, no colour, coloured solution not present (1)
no compounds

(b) **(i)** white ppt (1)

(ii) soluble (1)

(c) **(i)** white ppt (1)

(ii) insoluble (1)

(d) aq. NaOH (1) A*l* foil (1) warm (1)
ammonia (1) or gas turns litmus blue
or brown ring test: conc (1) H_2SO_4 (1) $FeSO_4$ (1) brown ring (1)

A*l*$(NO_3)_3$ (1) [10]

12 **(a)** **(i)** 26, 35, 47, 60 all correct (2) one error (1)

(ii) all points plotted correctly (1) smooth curve through points (1)

(iii) 60 s (1)

(iv) 32 °C (1)

(b) **(i)** all points plotted correctly (1) smooth curve through points (1)

(ii) 132 s (1)

(iii) 30 °C on graph 1 ------ 60 seconds (1)
60 seconds on graph 2 ------ 0.052 mol/dm^3 (1)
(Correct answer, 0.052 mol/dm^3 (2))

[For answers (a)(iii) and (iv), (b)(ii) and (iii) please read candidate's graphs to ± half small square.] [11]

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